

T Tests

LLE – Mathematics and Statistics Skills

This workshop will help you to understand hypothesis testing for one- and two-tailed tests, form the null and alternative hypotheses, and test them. It also looks at carrying out t-tests, you will need access to critical values of the t-statistic tables.

Useful Formulae

Pooled Variance, assuming equal variances:

$$s_{\text{pooled}}^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

One-sample t-test:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Two-sample t-test, assuming equal variances:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

1. Decide whether the following requires the use of a one-tailed or two-tailed test

- (a) To test whether there has been a change in the proportion of cats that prefer a cat food brand

Solution

Change indicates preference could increase or decrease, so two-tailed

- (b) To test whether there is a difference in shot accuracy of right-handed vs left-handed pool players

Solution

Difference indicates preference could increase or decrease, so two-tailed

- (c) To test whether a new road safety scheme has decreased the number of road accidents

Solution

Decreased number of accidents only, so one-tailed

2. A farmer test whether the mass of their strawberries is increased by using some magic growth formula. The known mean of the farmer's strawberries is 12 g each. A sample of strawberry plants is given the magic formula and a one-sample t-test is then carried out to see if the mean of the sample gives evidence of the formula working.

- (a) Write down the null hypothesis of the one-sample t-test

Solution

Null: The mean mass of the strawberries is 12 g

$$H_0 : \mu = 12$$

- (b) Write down the alternative hypothesis of the one-sample t-test

Solution

Alternative: The mean mass of the strawberries is greater than 12 g

$$H_1 : \mu > 12$$

- (c) The test does not find evidence that the formula increases the size of the strawberries, and so the farmer disposes of the remaining formula (in an environmentally friendly way). It becomes clear that the magic formula indeed works. Is this an example of a type I or type II error in the statistical test?

Solution

In this example the null hypothesis was not rejected when the null hypothesis was false. Therefore, it is a Type II Error

3. A lecturer believes that the mean IQ of the students in this year is higher than the mean value of 100. She gives a sample of the students an IQ test and gets a sample mean of 111. A one-sample t-test is then performed to see if this sample mean shows evidence at the 5% level of a mean greater than 100. The test gives $p = 0.0477$.

- (a) What are the null and alternative hypotheses?

Solution

$$H_0 : \mu = 100$$

$$H_1 : \mu > 100$$

- (b) Is there evidence, at the 5% level, of there being a higher than 100 mean IQ score?

Solution

The p-value is less than 5% (4.77%). Therefore, we would reject the null hypothesis at the 5% level, and state there is evidence of mean IQ being greater than 100.

- (c) Is there evidence, at the 1% level?

Solution

The p-value is not less than 1%. We do not have evidence at the 1% level.

4. A local council believes that a road safety campaign has reduced the number of traffic incidents on a particular stretch of road. Historically the mean number of incidents per week was 9.

- (a) Write down the null and alternative hypotheses

Solution

$$H_0 : \mu = 9$$

$$H_1 : \mu < 9$$

- (b) Over a 10-week period, after the road safety campaign, the mean number of accidents is recorded at 7.3 per week, with a sample variance of 3.2. Test the hypothesis, using a one-sample t-test, that you have set up at the 5% level.

Solution

$$\bar{x} = 7.3, \mu = 9, s = \sqrt{3.2}, n = 10$$

$$t = \frac{7.3-9}{\sqrt{\frac{3.2}{10}}} = -3$$

Look up the critical value on the one-tailed t-table, at the 5% level (0.05) and 9 degrees of freedom

$$t_{\text{crit}} = 1.833$$

Since 3 exceeds the critical value, we will reject the null hypothesis.

There is evidence that the scheme has reduced the number of accidents.

5. Samples of worms from two different sites are collected. Their lengths are measured before returning them to the wild. A two-sample t-test, assuming equal variances, is carried out to test for a difference in worm length between the two sites. The test was carried out using Microsoft Excel and the following was generated:

t-test 2-sample assume equal variances		
Length of Worm	Site 1	Site 2
Mean	50.9	48.2
Variance	4.999	9.874
Observations	10	14
Pooled Variance	7.88	
df	22	
t	2.31	
P(T<=t) one-tail	0.015	
P(T<=t) two-tail	0.031	

- (a) How big was the sample of worms taken from site 1 and from site 2?

Solution

$$n_1 = 10, n_2 = 14$$

- (b) Verify, showing your calculations, values for the:

- i. degrees of freedom

Solution

$$df = 10 + 14 - 2 = 22$$

- ii. pooled variance

Solution

$$\frac{(10-1) \times 4.999 + (14-1) \times 9.874}{10+14-2} = 7.88$$

- iii. t statistic

Solution

$$t = \frac{50.9 - 48.2}{\sqrt{7.88} \sqrt{\left(\frac{1}{10} + \frac{1}{14}\right)}} = 2.32$$

- (c) Given that the test was to look for a difference in worm lengths between the two sites:

- i. Write down the null and alternative hypotheses

Solution

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

- ii. Write down the p-value associated with the test

Solution

$$p = 0.031$$

- iii. Write a conclusion

Solution

There is evidence of a difference of lengths of worms, at the 5% level, between the two sites.

6. The waiting time (in minutes) for two customer service departments are given:

Department X: 1, 2.2, 1.5, 1, 1.7

Department Y: 1.1, 0.3, 0.8, 0.4, 0.8, 1.4, 1.1

Test whether there is evidence that the two departments differ in the waiting times, at the 5% level. You are given that the sample variance of department X is 0.257 and of department Y is 0.156. Equal variances can be assumed.

Solution

$$\bar{x} = \frac{1+2.2+1.5+1+1.7}{5} = 1.48$$

$$\bar{y} = 0.843$$

$$s_p^2 = \frac{4 \times 0.257 + 6 \times 0.156}{10} = 0.1964$$

$$t = \frac{1.48 - 0.843}{\sqrt{0.1964} \sqrt{\left(\frac{1}{5} + \frac{1}{7}\right)}} = 2.455$$

$$df = 5 + 7 - 2 = 10$$

Using the t-table, the critical value for the two-tailed test is 2.228. Since this t-value exceeds the critical value we can reject the null hypothesis. There is evidence of a difference in waiting times.